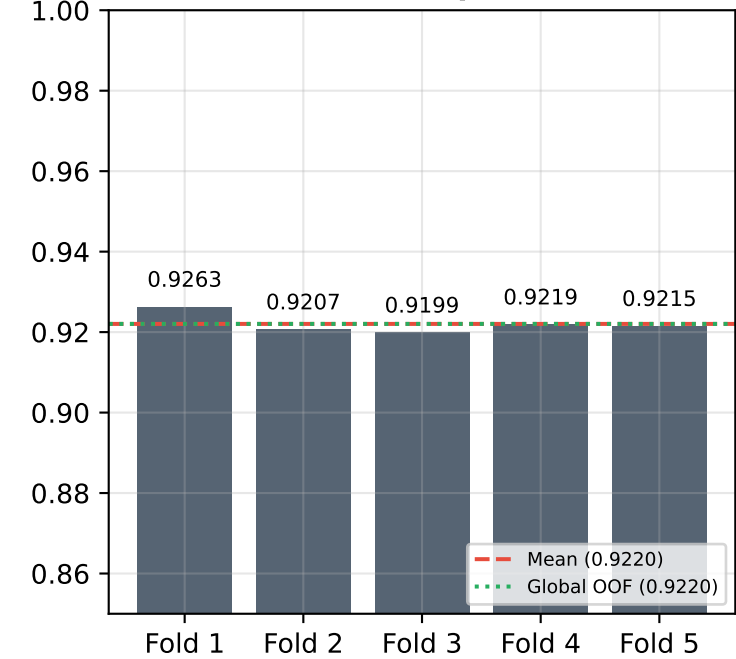
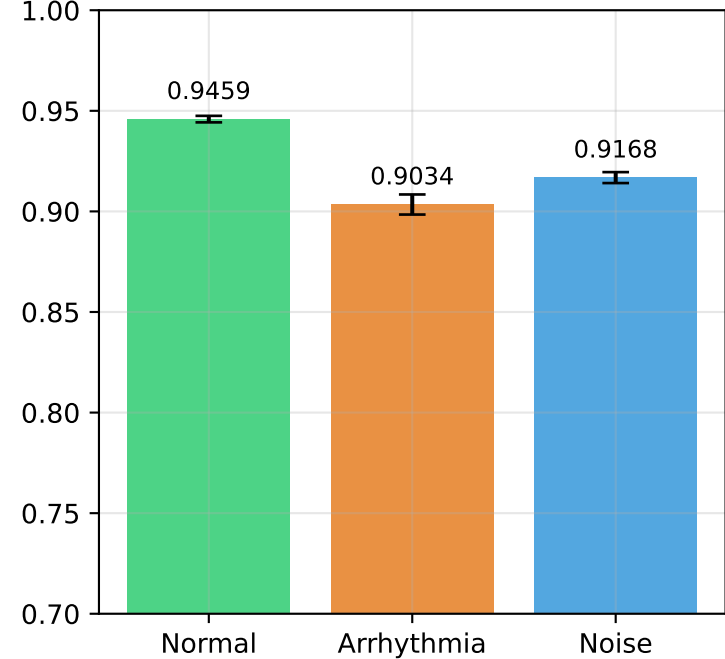


Composite Summary — ECGNet-SE-BiGRU Cross-Validation and Test Performance

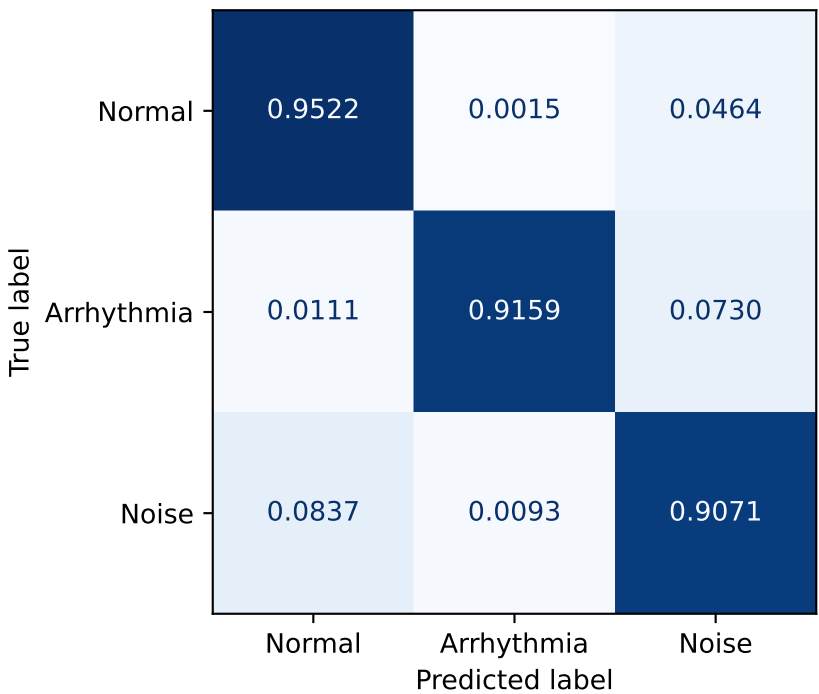
A. F1 Macro per Fold



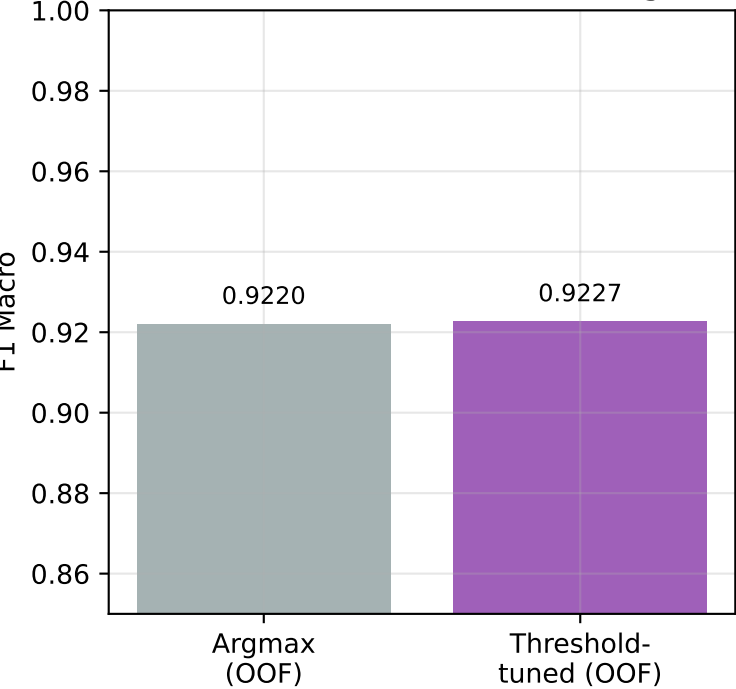
B. Mean Per-Class F1 ($\pm$ SD)



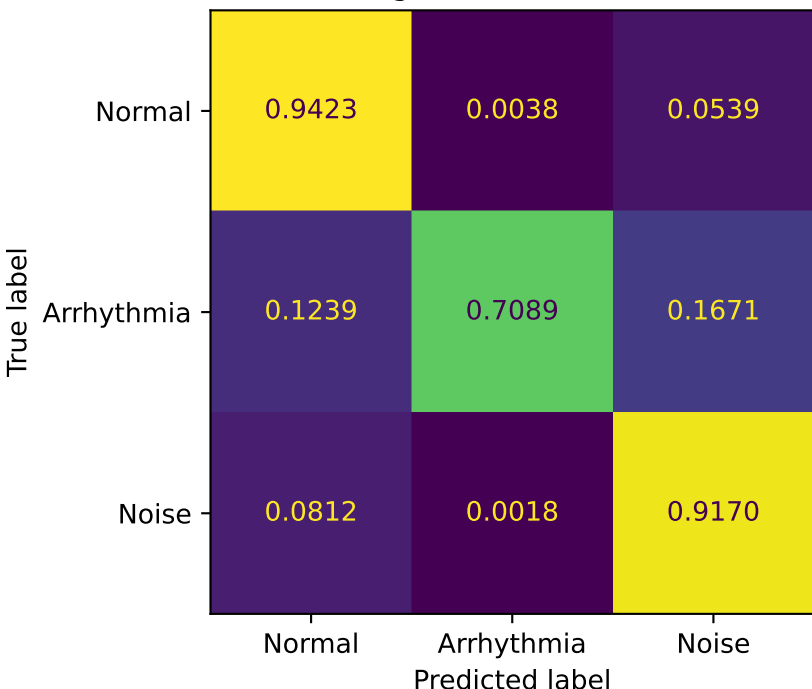
C. Global OOF Confusion Matrix



D. Effect of Threshold Tuning



E. External Test Confusion Matrix
Weighted F1 = 0.9306



F. Summary Metrics

Key Results Summary

CV Macro F1 (mean $\pm$ SD):
0.9220 $\pm$ 0.0022

Global OOF Macro F1:
0.9220

OOF Macro F1 (threshold-tuned):
0.9227

Optimal thresholds (N / A / Noise):
0.2857 / 0.4571 / 0.3286

External test Weighted F1:
0.9306